THE PRODUCT OF TWO MORPHISMS REPRESENTABLE BY ALGEBRAIC SPACES IS REPRESENTABLE BY ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. The factorization and base-change identifications are carried out in-span. The two same-chapter inputs are not carded and are declared external.

Allowed standard prerequisites (editorial).

- Composition stability. lemma-composition-representable-by-spaces (same chapter, NOT carded, declared external). Composes the two base changes.
- Base-change stability. lemma-base-change-representable-by-spaces (same chapter, NOT carded, declared external). Keeps each factor representable by algebraic spaces.

Source reference. Original lemma 0302: sources/stacks-algebraic/pinned-original-source.tex, source lines 876–876. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0301: sources/stacks-algebraic/pinned-original-source.tex, source lines 969–969. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}_i, \mathcal{Y}_i$ be categories fibred in groupoids over $(Sch/S)_{fppf}$, $i = 1, 2$. Let $f_i : \mathcal{X}_i \to \mathcal{Y}_i$, $i = 1, 2$ be 1-morphisms representable by algebraic spaces. Then*

$$f_1 \times f_2 : \mathcal{X}_1 \times \mathcal{X}_2 \longrightarrow \mathcal{Y}_1 \times \mathcal{Y}_2$$

is a 1-morphism representable by algebraic spaces.

Proof. Write $f_1 \times f_2$ as the composition $\mathcal{X}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1 \times \mathcal{Y}_2$. The first arrow is the base change of f_1 by the map $\mathcal{Y}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1$, and the second arrow is the base change of f_2 by the map $\mathcal{Y}_1 \times \mathcal{Y}_2 \to \mathcal{Y}_2$. Hence this lemma is a formal consequence of Lemmas 0301 and 0302. □

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.